


Part 1: Individual decision-making

F.S. task is to understand d.m. at the individual level
i.e. the level of economic agent or person makes

chapter 1: Decision making in general

But we our scope to economic
agent operating in a market economy

chapter 2-3 or the consumer theory given price and income
what type of consumer decisions do we expect

chapter 2. consumer choice

start by describing some elements of the prob.

1st there're fix # of goods and finite
of goods on commodities

Let x_i denote the quantity of good i ;
that is consumer $i = 1, 2, \dots, L$.

The bundle that the agent is consumed is denoted
 $x = (x_1, \dots, x_L)$

x is a vector, is supposed to be a col vect.

sec. the consumption set X Denote the set of possible bundles
we assume quantities are non-negative or
 $x_i \geq 0 \quad i=1, 2, 3$. we imagine quantities are
continuous $X = \mathbb{R}_+^L$

third, the agent consumers live in market econ.

Let w denote wealth P_i denote good

$i=1, 2, \dots, L$ Let $P = (P_1, \dots, P_L)$ assume prices
are all ~~posi~~ positive

$$P_i > 0 \quad P \cdot x = \sum_{i=1}^L P_i x_i \leq w$$

The budget set is the set of affordable or feasible products denoted B .

$$B_{p,w} = \{x \in \mathbb{R}_+^L : p \cdot x \leq w\}$$

The subset of bundles where the entire income is spent is $\{x \in \mathbb{R}_+^L : p \cdot x = w\}$

The budget line $L=2$ on the budget hyperplane.

properties of the budget set

- Given $w \geq 0$, B is nonempty since $x=0$ is affordable

- Given $p \succ 0$, B is bounded.

Limit to what can be afforded B is closed set.

B is compact set.

B is a convex set. If $x, x' \in B$, then

$$Tx + (1-T)x' \in B. \quad T \in [0, 1]$$

We have described fiscal envor. We can proceed in two ways

- the classical approach. START from Axioms on preferences and see what predic follow.

reveal pref. approach. start from choice
observable data.

something about the consumers from these choices